Analyzing Goodreads

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Introduction

- Our team analyzed the 2017 UCSD Goodreads dataset.
- This data was scraped from public user data on the Goodreads.com website
- The dataset contains three groups of data: meta-data of books, public shelves, and users' detailed book-reviews.
- Contains over 2,360,655 books (400,390 book series)
- 876,145 users (112,131,203 reads and 104,551,549 ratings)

Discovery Questions

- DQ1: What are the hidden associations between specific book genres and user-defined "shelves"? (Do readers of "Romantasy" (Romance Fantasy) also show high interest in other genres like fantasy?
- DQ2: Can we identify distinct Reader Personas based on interaction metrics? By clustering users based on their average rating variance, review frequency, and review length our goal is to discover if it is possible to classify readers into different groups?
- DQ3: What type of patterns emerge from highly-voted reviews compared to low-voted ones?

Discovered Patterns

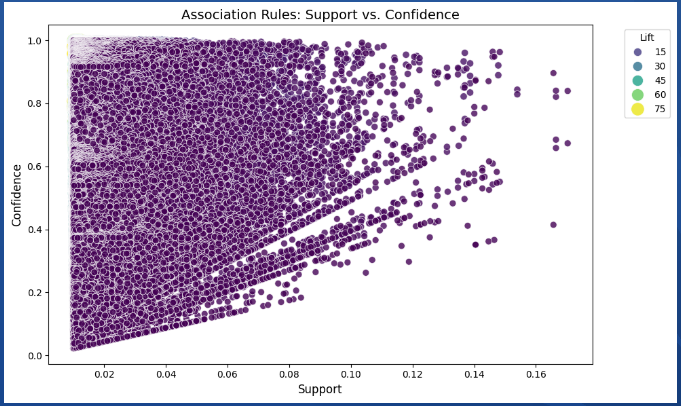
- DQ1: The FP-Growth algorithm did not find any strong association rules linking “Romantasy” or other related genres. We assume that readers who tag books with “Romantasy” treat it as a standalone category only a descriptive category
- DQ2: We were able to group readers into 4 different groups: Harsh consumer, Generous consumer, Powered reader, and standard reviewer
- DQ3: Analyzed engagement with reviews and relevance in the community. We discovered that longer reviews and star ratings do not correlate with higher community engagement

Approaches used

- FP-Growth algorithm: Treating user's shelves as transaction to discover rules with high-lift and confidence to identify relationships between genres
- K-means clustering: Used to answer DQ2. Apply on standardized numerical features
- Latent Dirichlet Allocation (LDA): Allow us to discover patterns within user's reviews and try to find community engagement

Discovery Question 1

- Our initial hypothesis for the first discovery question was proven to be false. The FP-Growth algorithm we created did not find any strong association rules linking Romantasy genres to other categories.
- Readers do not typically engage in cross-pollination and only use it as a descriptive tag.



Discovery Question 2

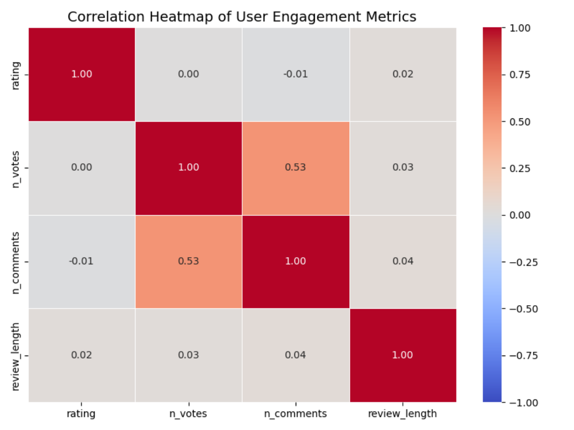
TABLE III
USER PERSONA PROFILES BY K-MEANS CLUSTER

Cluster	User Count	Review Frequency	Avg. Rating	Rating Variance
0	162270	9.32	2.93	4.62
1	94184	17.01	4.44	0.56
2	4595	321.72	3.78	1.09
3	80856	24.65	3.44	1.47

- A K-Means analysis was created to group readers together in four different groups based off their activity on the site.
- Cluster 0 Harsh Consumer: Third largest reading group but group with least amount of read books. Usually finds a 5 star book or more likely hates what they read
- Cluster 1 Generous Consumer: Biggest reading group; Only rates books occasionally but rates books highly and rarely rates them any lower.
- Cluster 2 Powered reader: Smallest reading group; Averages a review every single day and reviews the most books and firm in ratings
- Cluster 3 Standard reviewer: Second biggest group. Tend to be more active than cluster 1. Reviews a lot of books with little impact to the reader

Discovery Question 3

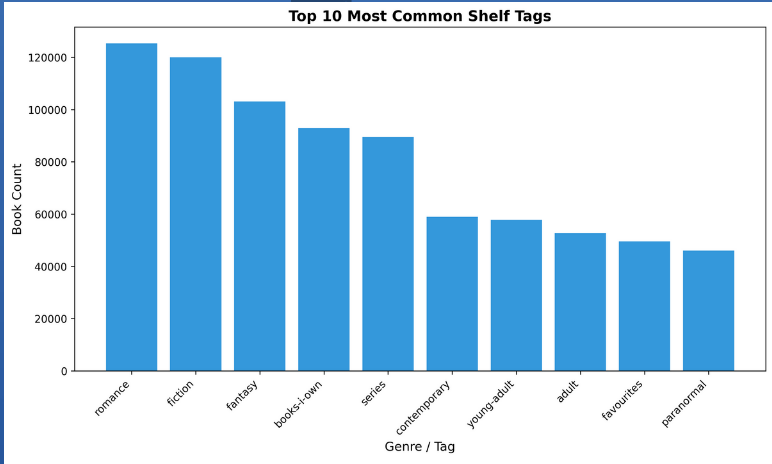
- Examined what patterns emerge from highly up-voted reviews compared to low engagement reviews.
- Our Pearson correlation analysis disproved our hypothesis our team discovered that while social interactions tended to scale together they were driven by factors other than the review
-

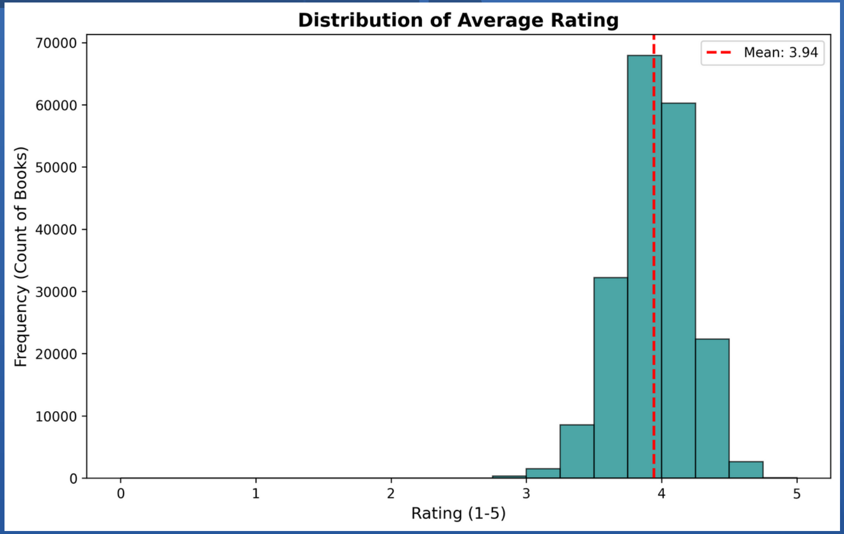


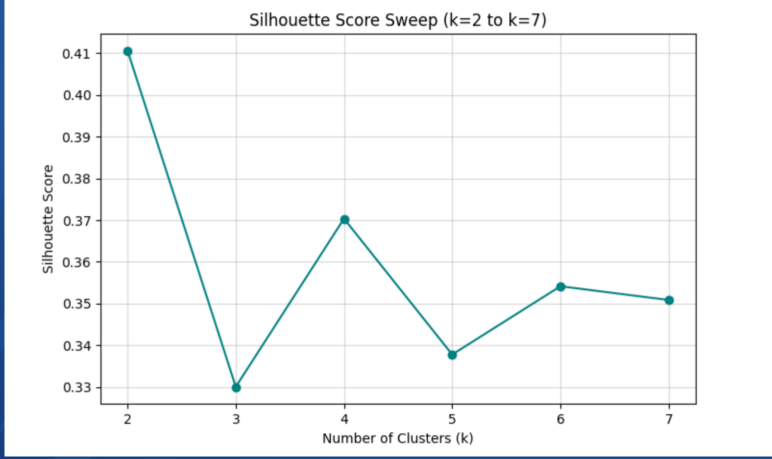
LDA Analysis

- As discussed for DQ3, our team could not find a correlation as to what drove user engagement so our team analyzed the reviews deeper using a Latent-Dirichlet Allocation
- We analyzed 3 distinct themes that drove community interaction on Goodreads
- Analytical Critique: Reviews focusing on world-building and pacing of the books. Suggests these reviews were popular and favored more
- Emotional Reactionaries: Reviews focused on feelings specifically mentioning “book hangovers” and crying after finishing. These readers loved the book so much they couldn’t get over it.
- Administrative / ARC disclaimers: Interesting one. Theses readers were mainly bloggers or disclosed that they received Advanced Reader Copies before the book launched

Results


Top 10 Most Common Shelf Tags
Book Count
0
20000
40000
60000
80000
100000
120000
romance
fiction
fantasy
books-i-own
series
contemporary
young-adult
adult
favourites
paranormal
Genre / Tag


Distribution of Average Rating
Mean: 3.94
Frequency (Count of Books)
0
10000
20000
30000
40000
50000
60000
70000
0
1
2
3
4
5
Rating (1-5)


Silhouette Score Sweep (k=2 to k=7)
Silhouette Score
0.33
0.34
0.35
0.36
0.37
0.38
0.39
0.40
0.41
2
3
4
5
6
7
Number of Clusters (k)

Validity

- Our team relied on the Elbow method and silhouette scores of 0.50 to show that the 4 clusters of reader groups found during the K-Means analysis are statistically significant
- Our LDA analysis explains our findings from our Pearson corelation tests

Limitations

- The UCSD Dataset was from 2017 so while it is recent enough to get a good picture of the current landscape
- 2017 was on the tail end of the “YA Dystopian craze”
- The 2020 Pandemic created a bunch of new readers and exploded the popularity of “Enemies to Lovers” and other Fantasy romance genres.
- If there was another dataset created our team expects our results to be significantly different

Conclusion

- Our team successfully analyzed the 2017 UCSD Goodreads dataset that examined the different type of readers and user interactions on the site.
- Our team were able to describe different types of readers and what drove them to reviews.
- We suspect that authors could use our work in help writing their own books to make their own books more successful.

Questions

- How confident am I that these patterns are real?
- What would I do differently with more time or data?
  - Find a newer dataset, the 2020 Pandemic saw a increase in readers and massive popularity boom for Fantasy and “Romantasy”. We suspect our results would vary significantly if there was a dateset made around 2021.
- How confident am I that this pattern is real and not an artificial?
  - Size of the dataset, booktubers, etc...